

Differentiating inverse trig functions

Basic skills

$$Q1. \frac{d}{dx} (3 \sin 2x) = 6 \cos 2x$$

$$Q2. \sin \theta = \frac{2}{3} \quad \cos^2 \theta + \sin^2 \theta = 1$$

$$\text{as } \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$

$$Q3. \sec \theta = 5 \quad 1 + \tan^2 \theta = \sec^2 \theta$$

$$\text{as } \tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{25 - 1} = \sqrt{24} = 2\sqrt{6}$$

$$Q4. y = \cos^{-1} x$$

$$a) x = \cos y \Rightarrow \frac{dx}{dy} = -\sin y$$

$$b) \frac{dy}{dx} = \frac{-1}{\sin y} = \frac{-1}{\sqrt{1 - \cos^2 y}} = \frac{-1}{\sqrt{1 - x^2}}$$

$$c) \frac{dy}{dx} (0) = \frac{-1}{\sqrt{1 - 0^2}} = -1$$

Q5. $\frac{dy}{dx}$ with $y = \tan^{-1}x \Rightarrow x = \tan y$

so $\frac{dx}{dy} = \sec^2 y \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y}$

$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + x^2}$

$1 + \tan^2 y = \sec^2 y$

Q6. $\frac{dy}{dx}$ with $y = \sec^{-1}x \Rightarrow x = \sec y$

$x = \frac{1}{\cos y} \Rightarrow \frac{dx}{dy} = \frac{-\sin y}{\cos^2 y} = -\sec y \tan y$

could recall this $\downarrow$

Quotient $\rightarrow$

so $\frac{dy}{dx} = \frac{-1}{\sec y \tan y} = \frac{-1}{x \sqrt{x^2 - 1}}$

$1 + \tan^2 y = \sec^2 y$

Q7. a) $\frac{dy}{dx}$, $y = \arcsin\left(\frac{x}{a}\right)$ so $x = a \sin y$

$\Rightarrow \frac{dx}{dy} = a \cos y \Rightarrow \frac{dy}{dx} = \frac{1}{a \cos y} = \frac{1}{a \sqrt{1 - \sin^2 y}}$

$= \frac{1}{a \sqrt{1 - \left(\frac{x}{a}\right)^2}} = \frac{1}{\sqrt{a^2 - x^2}}$

$$b) \frac{d}{dx} \left(\arcsin \left(\frac{x}{a} \right) \right) = \frac{d}{du} \left(\arcsin(u) \right) \frac{d}{dx} \left(\frac{x}{a} \right)$$

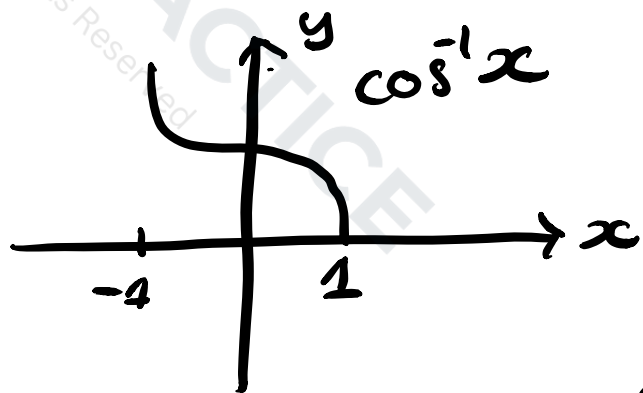
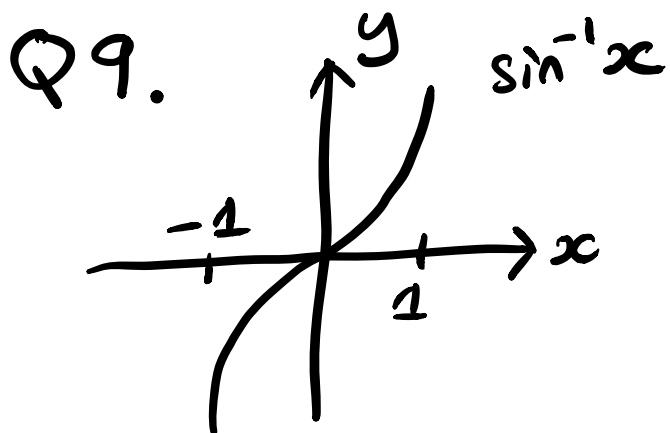
$$= \frac{1}{\sqrt{1-u^2}} \times \frac{1}{a} = \frac{1}{a\sqrt{1-\left(\frac{x}{a}\right)^2}} = \frac{1}{\sqrt{a^2-x^2}}$$

Q8. $y = \arctan 2x \Rightarrow x = \frac{1}{2} \tan y$

so $\frac{dx}{dy} = \frac{1}{2} \sec^2 y = \frac{1}{2} \sqrt{1+\tan^2 y} = \frac{1}{2} \sqrt{1+(2x)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{2}{\sqrt{1+4x^2}}$ so at origin $(0,0)$

$\frac{dy}{dx}(0) = \frac{2}{\sqrt{1}} = 2$ so $y = 2x$



Gradient of $y = \sin^{-1} x$ is always positive
 Gradient of $y = \cos^{-1} x$ is always negative
 so student is wrong

Q10. $\frac{dy}{dx}$

$$y = \cot^{-1} x^2$$

$$\Rightarrow x = \sqrt{\cot y} = (\cot y)^{\frac{1}{2}}$$

$$\frac{dx}{dy} = \frac{1}{2} \times \frac{1}{\sqrt{\cot y}} \times -\operatorname{cosec}^2 y$$

Sub $u = \cot y$
 $\frac{d}{dy}(\cot y) = -\operatorname{cosec}^2 y$

$$= \frac{-\operatorname{cosec}^2 y}{2\sqrt{\cot y}} = \frac{-\operatorname{cosec}^2 y}{2x}$$

$$= \frac{-(\cot^2 y + 1)}{2x} = \frac{-(x^2 + 1)}{2x}$$

$\operatorname{cosec}^2 y = \cot^2 y + 1$

so $\frac{dy}{dx} = \frac{-2x}{x^2 + 1}$

Q11. as $x = \sin y \Leftrightarrow \frac{1}{x} = \operatorname{cosec} y$

so $y = \sin^{-1} x \Leftrightarrow y = \operatorname{cosec}^{-1} \frac{1}{x}$

so $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$

so $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{d}{dx}(\sin^{-1} \frac{1}{x}) = \frac{d}{du}(\sin^{-1} u) \frac{d}{dx}(\frac{1}{x})$

$u = \frac{1}{x}$

$$= \frac{1}{\sqrt{1-u^2}} \times -\frac{1}{x^2} = \frac{1}{x^2 \sqrt{1-\frac{1}{x^2}}} = \frac{1}{x \sqrt{x^2-1}}$$

Q12. For $x = f(y)$

by implicit diff $1 = f'(y) \frac{dy}{dx}$

$$\text{so } \frac{dy}{dx} = \frac{1}{f'(y)} = \frac{1}{\frac{dx}{dy}}$$

Q13. $y = \arcsin\left(\frac{x}{3}\right)$

$$\begin{aligned} \text{a) } x &= 3 \sin y & \frac{dx}{dy} &= 3 \cos y = 3 \sqrt{1 - \sin^2 y} \\ & & &= 3 \sqrt{1 - \left(\frac{x}{3}\right)^2} = \sqrt{9 - x^2} \end{aligned}$$

$$\text{so } \frac{dy}{dx} = \frac{1}{\sqrt{9 - x^2}}$$

$$\text{b) } \int_1^2 \frac{2}{\sqrt{18 - 2x^2}} dx = \int_1^2 \frac{2}{\sqrt{2}} \times \frac{1}{\sqrt{9 - x^2}} dx$$

$$= \frac{2}{\sqrt{2}} \left[\arcsin\left(\frac{x}{3}\right) \right]_1^2 = \frac{2}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) - \arcsin\left(\frac{1}{3}\right) \right)$$

remember rad's

$$= 0.551$$